

Effects of Looming Tommy John Injury
on Pitch Performance in Major League Baseball

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ABSTRACT

This study aims to quantify the effect of imminent injury in Major League Baseball pitchers by comparing pitch characteristics in the 30 days preceding a UCL tear to those of similar healthy pitchers. Healthy control pitchers are evaluated at comparable points in their careers, defined by matching on career pitch count.

Propensity score matching is used to pair injured and healthy pitchers based on key covariates. Two-sided paired t -tests are then conducted to assess differences in pitch response metrics, specifically the mean and variance of fastball velocity and spin rate, within each pitcher's respective 30-day window.

No significant differences were detected between injured and control pitchers in the 30-day matched comparisons. However, results indicate that pitchers with higher career spin rates and greater spin rate variability are, on average, more likely to experience UCL injury.

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1 Introduction

Baseball was the first major sport to undergo an analytical revolution, largely due to the sport’s measurability and repetitive structure (Lewis, 2003). Much of this movement can be traced to the work of Bill James, who began publishing statistical analyses of baseball performance in the late 1970s and challenged traditional measures of player value. James and other early analysts demonstrated that commonly used statistics such as batting average failed to fully capture a player’s contribution to winning (James, 1988). Roughly two decades ago, these ideas began to gain widespread acceptance among Major League Baseball decision-makers (Lewis, 2003; Albert and Bennett, 2001), ushering in the modern era of baseball analytics. Advanced statistics, commonly referred to as sabermetrics, are now widely used throughout Major League Baseball to inform decision-making in player evaluation, roster construction, and game strategy (Cho and Jin, 2026; Baumer et al., 2014; Mizels et al., 2022; Morgan and Ravindran, 2017; Wu et al., 2025). By 2012, each MLB team had hired “at least one sabermetrician” (Encyclopaedia Britannica, nd). While great strides have already been made, there are still many current public and private research projects aimed at understanding the game better, often centered around the most consistent part of a baseball game – pitching.

Starting pitchers are increasingly susceptible to major elbow injuries requiring Tommy John surgery (Camp et al., 2018), named after pitcher Tommy John, who was the first Major League player to successfully undergo the surgery in 1974. Tommy John surgery is a procedure used to repair or replace the damaged ulnar collateral ligament (UCL) in the elbow, an injury often caused by the extreme stress of repetitive, high-velocity throwing motions. This trend raises important questions: why are these injuries becoming more common, and can they be prevented?

Pitching is the central component of every baseball game. While teams pursue strategic advantages such as base stealing or advancing runners on sacrifice flies, none are as influential as pitching, since every play begins with a pitch. Baseball analysts and executives consistently identify pitching as the most important factor in winning games, accounting for roughly 60% of team success in expert evaluations (Skipper, 1980). This means it is difficult for a team to win with poor pitching, yet acquiring and maintaining multiple high-quality starting pitchers is costly and difficult. Teams pay millions per pitcher and continually grapple with how to properly value their performance in the labor market (Ludlow, 2023; Krautmann et al., 2009). Consequently, each pitch and outing carries substantial value. Injuries to pitchers’ elbows can significantly hinder a team’s chances of success, as injured pitchers may miss extended time or perform below their usual level. Identifying when an elbow injury may be imminent would allow teams to limit a pitcher’s workload, protecting the pitcher’s health while also preventing declines in performance.

In addition to benefiting professional teams, research on pitcher injuries is valuable to the broader public, as millions of individuals participate in baseball each year, with recent estimates (based on Sports & Fitness Industry Association

data) exceeding 16 million participants in the United States (Major League Baseball, 2023).

This study uses propensity score matching to pair injured pitchers with comparable healthy pitchers in order to examine whether significant differences in pitch characteristics emerge prior to an injury requiring Tommy John surgery. Rather than attempting to predict the timing of injuries, the goal is to identify meaningful relationships between specific pitch metrics and the occurrence of injury.

2 Literature Review

Many studies use existing data, provided by MLB (Camp et al., 2018; Pareek et al., 2021; Bushnell et al., 2010; Chalmers et al., 2016; Prodrromo et al., 2016; Mayo et al., 2021), and a few researchers have done their own experiments to evaluate the predictors of pitcher elbow injuries Fleisig et al. (1995); Slowik et al. (2019). There are some heuristics in the industry about when pitchers may be getting injured, and even some data that seems to indicate injury, but it is important to use only validated statistical methods to assess injury (Pareek et al., 2021). Overall, researchers are doing influential work, but the bulk of findings likely lie within private organizations. Because of this, there is room for advancement of pitching research in the public domain.

Studies focused more on biomechanics yield results about what may cause pitcher injuries, such as tension in biceps tendons (Fleisig et al., 1995). However, this type of data can only be found in controlled experiments or proprietary medical or team records. Therefore, this study focuses on using tabular, publicly available, game-measured statistics. They are easier and cheaper to use, and could be used in other leagues beyond MLB, since ball speed and spin rate are easier to measure than biokinetic characteristics.

Some studies focus on this widely available data, and the main conclusion is that higher normal/pre-injury pitch speeds are associated with UCL/elbow injuries (Chalmers et al., 2016; Prodrromo et al., 2016; Bushnell et al., 2010; Pareek et al., 2021). They also found that fastball velocity and spin rate decrease once injury sets in (fifteen days before the injury is announced), and pitchers may throw more curveballs, which is a slower pitch. (Mayo et al., 2021). The Mayo study is similar to this proposed study, but a time series “Mann-Kendall” trend test was used, which looks at changes within pitchers. This study relies on matching and finding differences between healthy and injured pitchers leading up to injury, as well as quantifying the difference in pitch metrics, rather than the strength of the relationship like in Mayo (Mayo et al., 2021).

The statistical technique used for comparing healthy pitchers to injured ones is propensity score matching. This method creates pairs of treated and untreated units with similar probabilities of treatment, enabling outcome comparisons that approximate those from randomized experiments (Austin, 2011). This technique is helpful because it can take in one to many ‘confounding’ variables into a propensity score model, and output a probability (Rosenbaum and Rubin,

1983), 0 to 1, that can easily be used to match treated and untreated pitchers.

3 Methodology

3.1 Structure

To accomplish the goals of the research paper, which is to determine the effects of being an “imminently injured” MLB pitcher, (being within 30 days of the last game before halting baseball activity because of a Tommy John injury), there are three main steps that must be completed (see Figure 1). The first step depicts that pitchers must be classified as injured or healthy, which are treated and control subjects, respectively. Next, injured pitchers must be matched to a control for pitch count, maximum pitch speed, and career average spin rate. This is accomplished using propensity scores. Propensity scoring is helpful because it gives us probabilities that can be used to match pitchers by pairing nearest neighbors (Austin, 2011). Finally, differences in pitch velocity and spin rate are compared using paired t-tests to determine if there is a difference in pitch outcomes in the treatment groups.

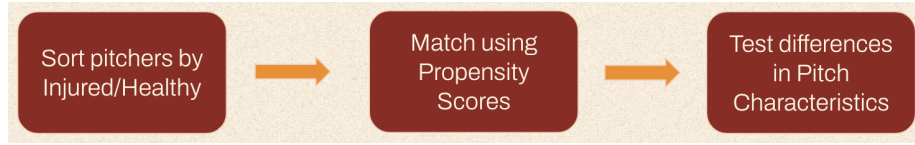


Figure 1: Research Process

3.2 Data

Pitch-by-pitch MLB data from Baseball Savant (Baseball Savant, nd), filtered for just four-seam fastball, dating 2010-2024 were used to calculate relevant averages and variances of pitch velocity and pitch spin rate. There exists a row in the data for every single pitch, then these rows are then grouped by pitcher, game, and pitch type, and summarized by mean velocity and spin rate, and career and game pitch count. In addition, pitch counts from 2008-2010 were used to provide accurate career pitch counts for players from 2010 forward. These pitch-by-pitch data were then joined to an injury data set, obtained by scraping tables from prosportstransactions.com (Pro Sports Transactions, nd). This injury information identifies if and when pitchers had Tommy John surgery. After filtering through all injuries, 135 pitchers were identified as having Tommy John surgery in the 2010-2024 window.

3.3 Pitch Metrics as Responses

Each pitcher’s average and variance in four-seam fastball pitch velocity and spin rate were calculated per game. Pitch velocity is measured in miles per

hour (MPH) and spin rate in revolutions per minute (RPM). Pitchers are incentivized to throw harder to make it harder for batters to hit, with averages usually lying in the 90’s. Spin rate has traditionally been emphasized in the analysis of breaking pitches due to its role in generating lateral movement; however, for fastballs, higher spin rates increase lift via the Magnus effect, resulting in reduced vertical drop, perceived “ride,” and increased deception to hitters (Nathan, 2008; Heilmann et al., 2022). This is when pitches do not fall towards the earth as fast, giving the illusion that the ball is rising, making it more difficult to hit.

In the data set, pitchers threw anywhere from single digit pitches up to 100+ pitches in a single game. A weighted mean was used, with pitch count per game as the weight, to calculate averages and variances within a given time period (either for a career, or a 30-day window).

For treated pitchers, pitch metrics were calculated using data from the 30 days prior to their final game appearance before undergoing Tommy John surgery. The condition was that pitches from these 30 days had to come from the same season as the injured pitcher’s injury. Many pitchers had zero pitches in this 30-day window, meaning they got injured sometime over the off-season. This study looks at the effects of being “imminently injured”, and since there is no way to know from the data when these off-season injuries occurred, these pitchers had to be removed from the dataset. This took the count of pitchers with Tommy John injuries from 135 to 83.

These 30-day pre-injury metrics were then compared to those of a matched control group, as described in the following section.

3.4 Matching

Each player was assigned a treatment: 1 if they ever had Tommy John surgery, and 0 if not. Each ‘treated’ player was then matched to two ‘control’ pitchers, based on maximum career fastball velocity and average fastball spin rate, using propensity scores. Pitchers’ fastest pitch speeds were used because they are more indicative of the strain a pitcher puts on their arm than their average fastball. Slowik et al. (2019) showed that higher velocities across different pitchers were not necessarily associated with higher elbow stress, but that higher pitch speeds within pitchers presented more risk to the elbow. Matching by maximum velocity aims to compare pitchers who have similar elbow-torque stress thresholds, so that they are comparable and representative of the proven ‘within pitcher’ effect. It is not a perfect solution, but it mitigates risk from only comparing random pitchers at times of injury. Bushnell et al. (2010) also showed maximum velocity to be significantly associated with elbow injuries.

Fastball spin rate was also included in the propensity score model, because it is a *possible* cause of injury, and helped balance the treatment and control groups, so that the difference in spin rate from control to injured group could be used later on. In some cases, controlling on an outcome variable would violate the causal framework, but since the outcome is a difference in a 30-day window, and not career spin rate, a slightly different spin rate during imminent injury

would not have a significant effect on career spin rate average.

The propensity score matching uses a generalized linear model with a binomial link function, with treatment (ever injured) explained by maximum pitch velocity in mph, and average spin rate in rpm.

Let T_i denote the treatment indicator for pitcher i , where $T_i = 1$ if the pitcher was ever injured and $T_i = 0$ otherwise. Let X_{i1} represent maximum pitch velocity, and X_{i2} average career spin rate. The propensity score is defined as

$$e(x_i) = P(T_i = 1 \mid X_{i1}, X_{i2}).$$

$$\log \left(\frac{P(T_i = 1 \mid x_i)}{1 - P(T_i = 1 \mid x_i)} \right) = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2},$$

where β_0 is the intercept, β_1 represents the change in the log-odds of injury associated with a one-unit increase in maximum pitch velocity, β_2 represents the change in the log-odds of injury associated with a one-unit increase in average career spin rate, x_{i1} denote the maximum pitch velocity (mph), and x_{i2} denote the average career spin rate (rpm).

The goal of propensity scores is to estimate the probability that a subject receives treatment, based on observed covariates that influence both treatment assignment and the response. In this context, maximum pitch velocity, which is known to be associated with increased injury risk, serves as a key predictor in estimating this probability. This allows for comparison between subjects with similar propensities to receive treatment, where one pitcher received the treatment and the other did not. Average spin rate is also included since it *may* have a relationship with causing injuries. The outcome effects for these matched pitcher comparisons are later used to estimate the effects of being “imminently injured”.

Once propensity scores were calculated, each treated pitcher was assigned the nearest two matches. This was done by searching for the two control pitchers’ propensity scores that had the lowest absolute difference from the treated pitcher’s score, and then matching each of those healthy pitchers to the treated pitcher. Then, the career pitch count at the time of injury was noted. The day when the career pitch count of the control pitcher most closely lined up with the time of its matched treatment’s injury was found, and the control “30-day window” was 30 days before that date (see this matching process in Figure 2). Then the control pitcher who had the larger absolute difference in pitch count was removed from the group. This was done because sometimes pitchers’ matches had not thrown as many career pitches as the treated pitcher had when he got injured. Having two potential matches and removing the worse match allowed for closer pitch count comparisons. Finally, the pitch statistics (mean and variance of fastball pitch speed and spin rate) in the 30-day windows were averaged, weighted by the number of pitchers per game.

3.5 Comparing Groups

Each matched pair consisted of one pitcher who underwent Tommy John surgery (treated) and one who did not (control). Pitchers were matched based on similar workload and fastball velocity capabilities, and pitch response metrics were calculated in 30-day windows at comparable pitch counts in their careers.

Four paired t -tests were conducted to compare the two groups, evaluating the average and variance of fastball velocity and fastball spin rate over the 30-day window preceding injury (or corresponding 30-day window for controls). These tests were used to determine whether significant differences exist in the 30 days prior to injury for pitchers who tore their UCL, and in a matched 30-day period for the control group.

Let

$$d_i = x_i^{(injured)} - x_i^{(control)}, \quad i = 1, \dots, n$$

The hypotheses for the paired t -test are

$$H_0 : \mu_d = 0$$

$$H_A : \mu_d \neq 0$$

where μ_d is the mean difference between matched pairs.

The test statistic is

$$t = \frac{\bar{d}}{s_d/\sqrt{n}}$$

where

$$\bar{d} = \frac{1}{n} \sum_{i=1}^n d_i$$

and

$$s_d = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (d_i - \bar{d})^2}.$$

Under H_0 , the statistic follows a t distribution with $n-1$ degrees of freedom.

4 Results

The results did not yield significant differences for any of the four response variables: average and variance of fastball velocity and fastball spin rate over the matched 30-day window prior to injury. As seen in Figure 3, the difference in spin rate variance is slightly positive but insignificant, meaning treated pitchers had no different spin rate variance in the treated vs control 30-day windows.

The difference in average spin rate, in Figure 4, was also negligible. Figures 5 and 6 indicate no significant difference in any pitch speed-related metrics for injured and healthy players.

There is a weak correlation (.115) between the difference in velocity and spin rate in this matched set of pitchers (Figure 7). This means that for a matched pair, greater difference in velocity for control or treated did not correlate much with greater difference in spin rate for control or treated. Additionally, there are two outliers in the top right corner that seem to account for much of the positive correlation. If those were removed, the correlation would be even closer to zero. Thus, changes in spin rate and velocity between a matched pair seemed to be essentially independent.

Something to note is the comparison between Tables 1 and 2. Table 2 shows the differences between the pitch metrics for the pitchers' whole careers, in each group. Ideally, each of the career tests would be insignificant, and the window tests would be significant, indicating a real treatment effect. However, this was not the case. An interesting note is that the career fastball spin rate averages and variances are significant (.012 for spin rate avg and .05 for spin rate variance). Lastly, the spin rate confidence intervals in the two tables do not overlap, suggesting that the difference between injured and control pitchers is larger when measured over their careers than in the 30-day window preceding injury. These findings will be discussed in the next section.

Variable	Point Estimate	95% Confidence Interval	p-value	Units
Velocity	-0.18	(-0.66, 0.31)	0.478	mph
Velocity Variance	0.14	(-0.06, 0.34)	0.170	mph
Spin Rate	13.6	(-23.3, 50.5)	0.464	rpm
Spin Rate Variance	-1131.2	(-3480.3, 1217.9)	0.340	rpm

Table 1: **Injury – Control, 30-day Window.** Estimated differences between pitchers within 30 days of injury and matched control pitchers with 95% confidence intervals.

Variable	Point Estimate	95% Confidence Interval	p-value	Units
Velocity	-0.35	(-0.86, 0.15)	0.169	mph
Velocity Variance	0.11	(-0.03, 0.26)	0.134	mph
Spin Rate	239.0	(53.8, 424.2)	0.012	rpm
Spin Rate Variance	1390.5	(-2.4, 2783.3)	0.050	rpm

Table 2: **Injury – Control, Career.** Estimated differences based on career pitch metrics with 95% confidence intervals.

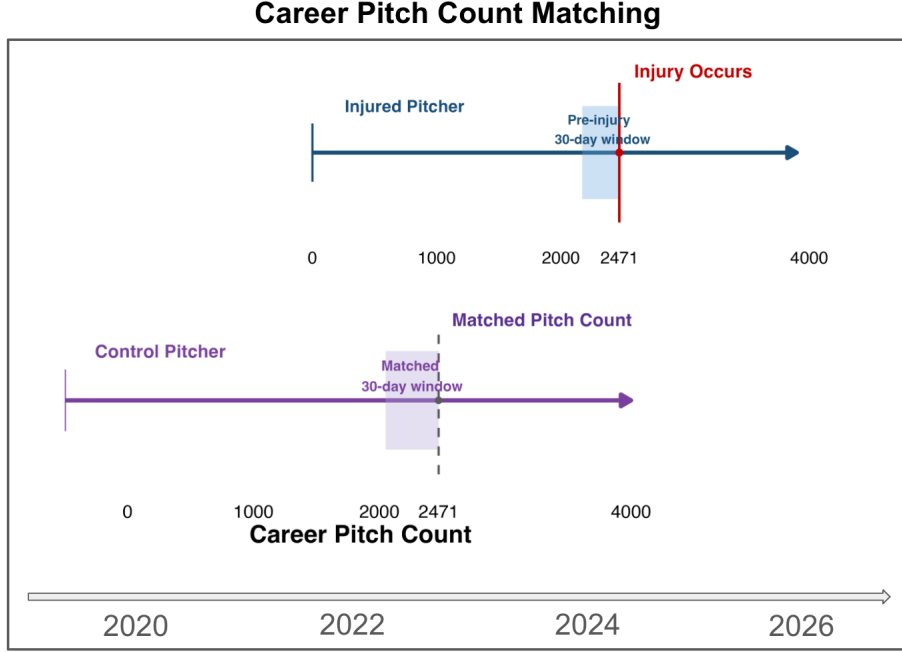


Figure 2: Visual of how career pitch counts were matched. Injured pitcher 30-day windows (the 30 days before injury) are tested against the 30-day window of healthy pitchers, at a time when they had a similar career pitch count.

5 Discussion

5.1 Limitations

There are considerations to be made when analyzing these results. The first is the assumptions - it is assumed that the only confounding variables that are related to the probability of receiving the treatment and the outcome pitch metrics were career maximum pitch velocity, career pitch count, and career spin rate average. In reality, there may be more, but from the literature, none have shown to be significant. Potentially useful variables to control for could be age and distribution of pitch type. Another assumption is that career pitch count is a true indicator of lifetime strain on an arm. These may be good estimators, but since access to pre-MLB pitch data is limited, it is not known how many pitches players threw in minor leagues, college, high school, etc. Since some pitchers skip college, and all stay in the minor leagues for different periods of time, they may be different ages when starting MLB and therefore have different lifetime pitch counts. However, using MLB career pitch count is still a justifiable way to control for pitch volume, since everyone is counted the same way, and it seems likely that true lifetime pitch count would be correlated with MLB pitch count.

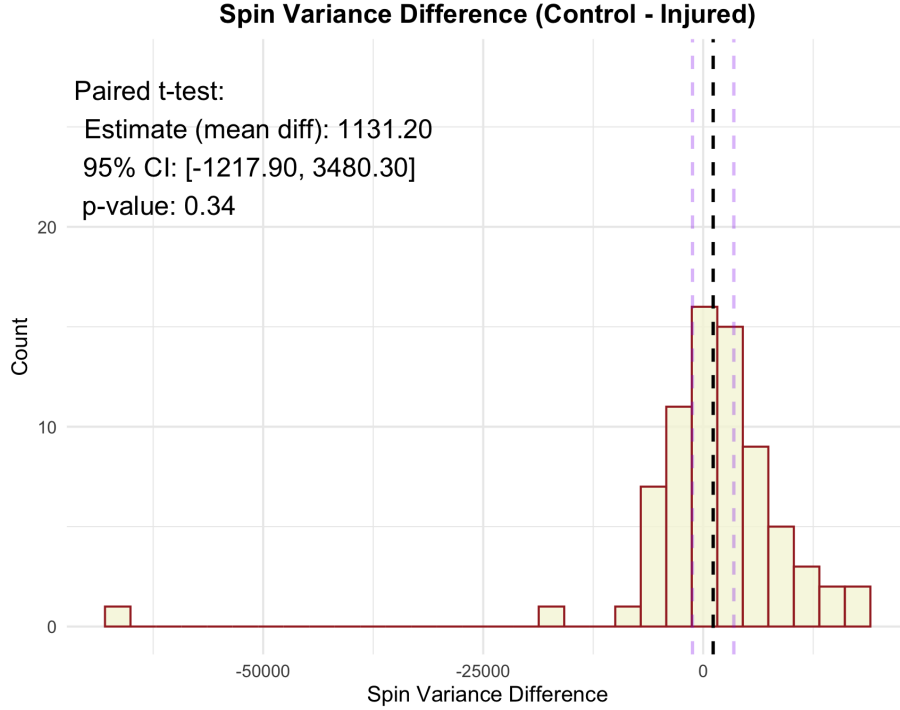


Figure 3: Difference in Spin Rate Variance

The second set of considerations to be made is that there were two thresholds used that may not be at the optimal level. The first is that a limit of 30 days prior to injury was chosen to determine an “imminently injured” period. A period of 30 days was used because it was long enough (see Figure 8) to have at least a few hundred samples for most players, but short enough that there could reasonably exist a sign of the body preparing for injury. If it were increased to more than 30 days prior to injury, there is less chance that pre-injury fatigue would have set in since it would be further away from the UCL tear. An injury study from (Mayo et al., 2021) uses 15 games prior to injury to track changes in pitch characteristics, but their approach aims to find a change over time in those outings, while this paper seeks to find a period of time where effects of being “imminently injured” have already set in, since they are compared to time periods from other healthy pitchers. In short, games from 30 days prior to injury may not be the ideal cutoff. Another underlying assumption with this 30-day threshold is that the fatigue, or effect of being imminently injured, was constant along the 30 days. This assumption may not be true, as the effect of being imminently injured may get larger in the days closer to the time of announced injury, but 1) if that is true, there should still be an increased average, and 2) for the scope of this methodology, this assumption fits well.

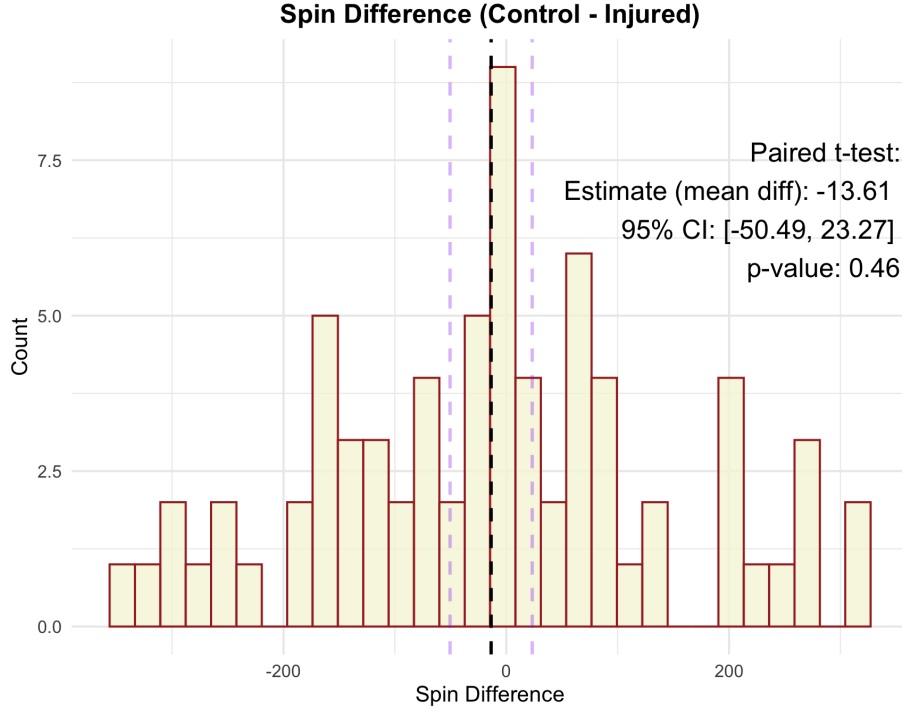


Figure 4: Difference in Mean Spin Rate

The other threshold-related concern is that each treated pitcher was matched with *two* control pitchers, before ultimately only keeping the match with the smallest career pitch count absolute difference. Increasing the number of matches could have improved the strength of matches in regards to having similar career pitch counts, but at the cost of finding the best match on maximum pitch velocity and spin rate. No optimization method was used to determine the number of matches.

5.2 Interpretation of Results

5.2.1 Changes in 30-day Window metrics

The results show that imminently injured pitchers do not have different fastball velocity or spin rates in the 30-day window prior to injury, on average, than their healthy counterparts at similar pitch counts. This means that there is no effect of being imminently injured on the pitch outcomes; so tracking changes in fastball speed and spin rate averages and variances would not be a helpful indicator for detecting injury.

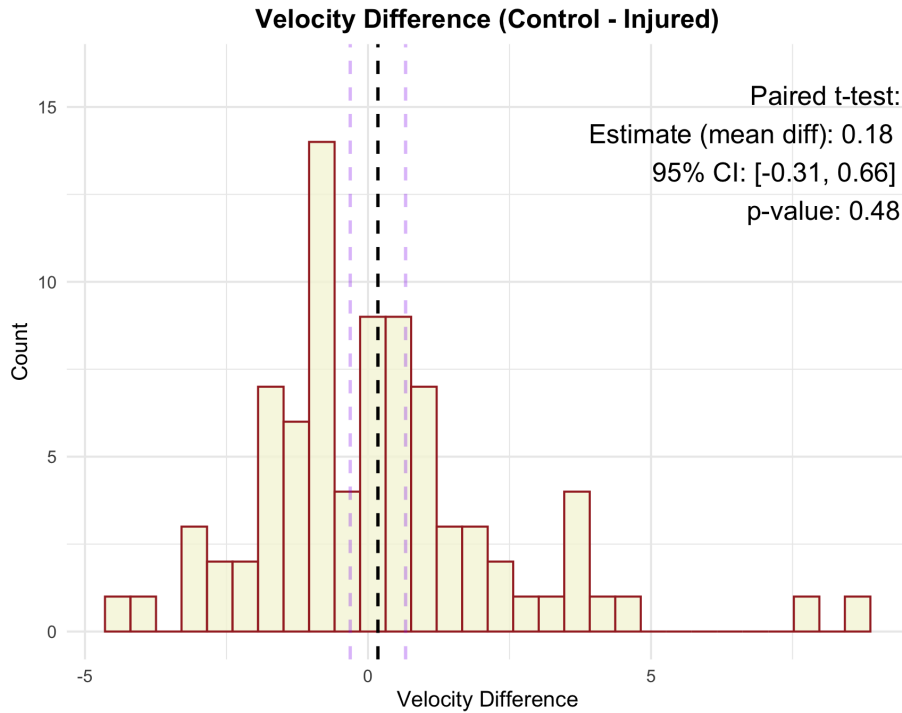


Figure 5: Difference in Mean Velocity within healthy/injured matched pairs

5.2.2 Changes in career Averages

Despite no significant differences observed in the spin rate average and variance of the 30-day window, there were significant differences in the *career* spin rate average and variance differences (Table 2). This shows that on average, pitchers who had higher career spin rates were more likely to be injured. Since average career spin rate was included as a covariate in the propensity score model, spin rates should have been balanced between the treatment and control groups. Since there was a significant difference, it is evidence that higher spin rate could be correlated with injury. Pitchers who squeeze the ball hard to get good ball rotation may be more injury prone over the course of a career, but spin rate does not appear to be an indicator of an imminent injury.

5.3 Application

Ignoring the 30-day window metrics, knowing that higher career spin rate is correlated with injuries could be helpful in drafting, trading away, and signing certain pitchers. If a team has a highly touted pitcher, but believes there is an injury risk due to spin rate, they could sell or trade that player. Additionally, teams could help develop pitchers to not spin the ball as hard to preserve their

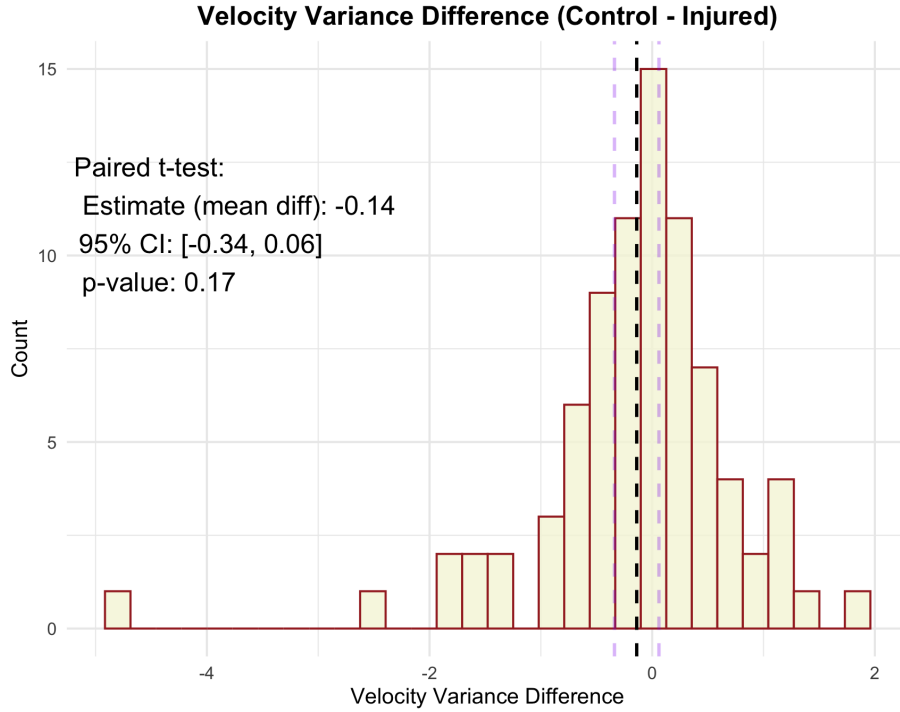


Figure 6: Difference in Pitch Velocity Variance

UCL.

6 Conclusion

In this matched-pairs analysis, where the metrics were taken within 30 days of injury, no significant differences were observed in the mean or variance of pitch velocity or spin rate for fastballs. There is evidence that pitchers who have higher career fastball spin rates, and have higher career variance in spin rate, are more apt to get injured in their career.

Future work might include testing arm angle as a response to the treatment. Arm angle is only publicly available from 2024 on, but there are efforts being made to retroactively predict arm angles based on ball location, height, and pitcher. It is possible that arm angle dips when the UCL becomes stressed, or that the arm angle slot is less consistent. If reliable pitch-by-pitch arm angle could be measured or calculated, its response to an imminent UCL injury could easily be tested given the framework already laid out in these research methods.

Secondly, there are other confounding variables that could be used as covariates in the propensity score model such as age, and pitch arsenal. Pitch arsenal is a distribution of the type of pitches a player uses. Age may be correlated with

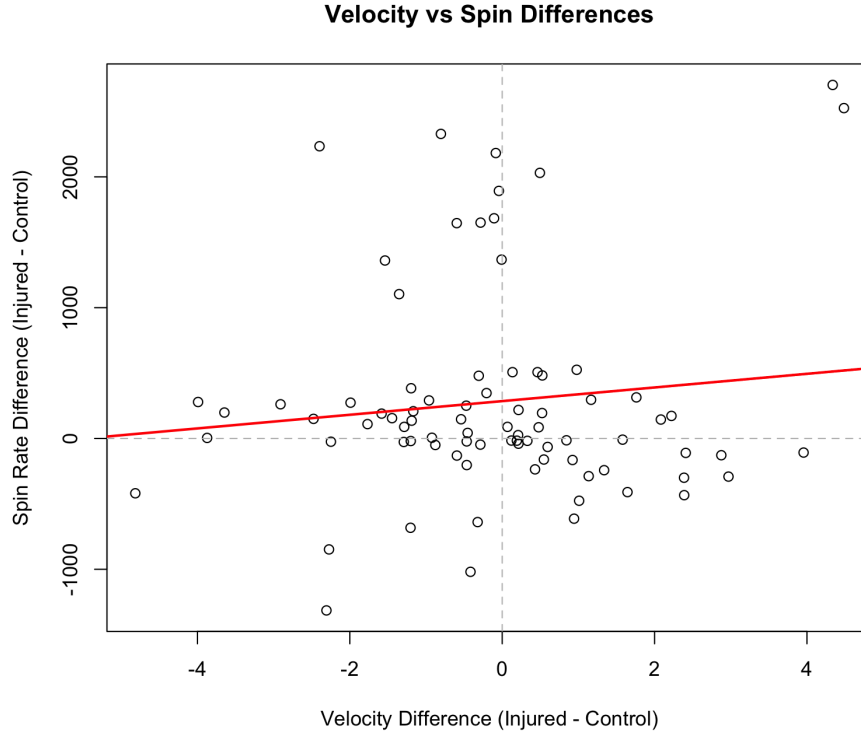


Figure 7: Differences in velocity v. Differences in Spin Rate. There is not a strong relationship.

career pitch count, but it could account for some of the missing pre-MLB data for players who spent significant time in college and the minor leagues. These factors could lead to better matches.

This research only looked at fastball metrics. Introducing breaking balls, where velocity is not emphasized as much but spin rate is rewarded, could lead to new findings. Another instance of not using the whole dataset is with injuries - only injuries resulting in Tommy John surgery that occurred during the season, were used (83). There are many more known injuries that could be analyzed as well, but methods would have to adapt to different types of treatments/injuries.

The assumptions, methods, and thresholds used in this study are not without limitations. With additional data, improved modeling approaches, and thresholds informed by biomechanics and sports science, further insight into injury risk among MLB pitchers could be achieved.

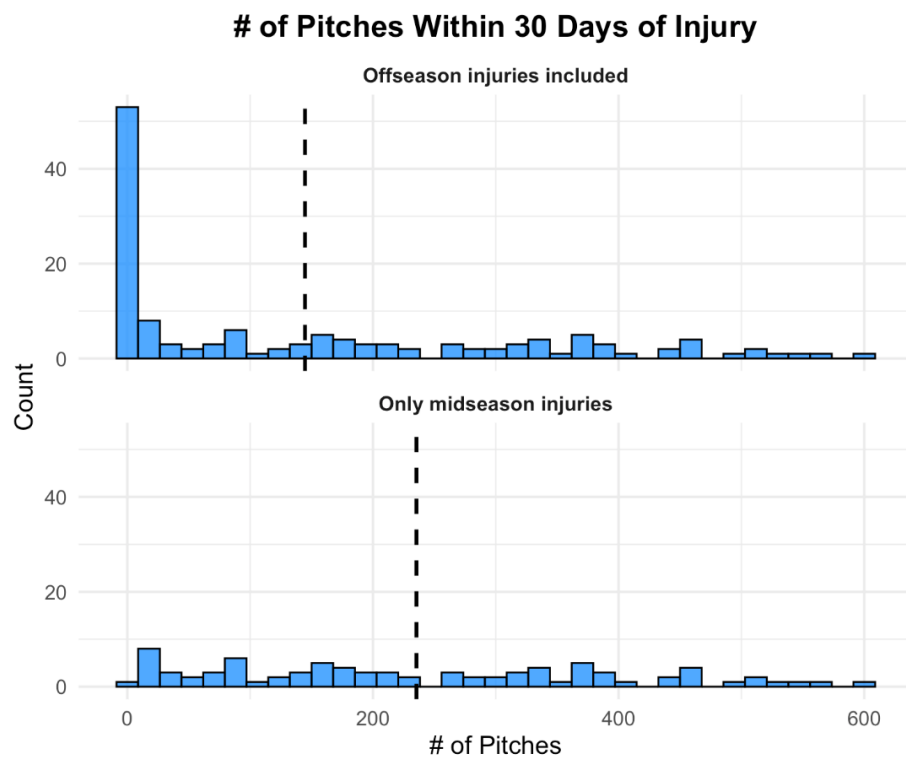


Figure 8: Number of Pitches Within 30 days of Injury

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